

Capita Experimental Results Report

Yiwen Hou

Abstract

This report compares **POD-RBF** and **cGAN-CNN** for thermal field prediction under limited-data conditions. Results show that **POD-RBF** remains the better solution in the current setting. The best POD-RBF configuration, `svd + rank=20 + thin_plate_spline`, achieves the highest average R^2 of **0.9265** and the strongest worst-case performance. Although cGAN improves after tuning and longer training, it still does not surpass the best POD-RBF result.

1. Experimental Overview

Two surrogate modeling pipelines are studied:

- **POD-RBF**: first reduces the thermal field by POD, then predicts reduced coefficients with RBF interpolation.
- **cGAN-CNN**: directly learns the mapping from input conditions to the output thermal field.

Three metrics are used: average R^2 , maximum R^2 , and minimum R^2 , representing overall accuracy, best-case performance, and worst-case robustness.

For POD-RBF, the main variables are `svd-method`, `rank`, and `rbf-kernel`. For cGAN, the main variables are learning rate, batch size, and adversarial weight.

2. Overall Model Comparison

Table 1 compares the baseline POD-RBF, tuned cGAN, and best POD-RBF result.

Table 1: Overall model comparison

Model	Configuration	Avg R^2	Max R^2	Min R^2
POD-RBF (baseline)	randomized_svd, auto kernel	0.9229	0.9880	0.7781
cGAN-CNN	epochs=2000, lr=1e-4, adv=0.01	0.9209	0.9880	0.7542
POD-RBF (best)	svd, rank=20, thin_plate_spline	0.9265	0.9856	0.8043

cGAN reaches a similar best-case result, but its average and minimum R^2 are lower. The optimized POD-RBF gives the best overall accuracy and stronger robustness.

3. POD-RBF Parameter Tuning

A total of 13 POD-related settings were compared. The best results are summarized in Table 2.

Table 2: POD-RBF tuning summary

Rank	Avg R^2	Configuration	Change
1	0.92649	svd + rank=20 + thin_plate_spline	+0.0036
2	0.92649	svd + rank=20 (auto kernel)	+0.0036
3	0.92570	svd + rank=15	+0.0028
4	0.92474	svd + rank=30	+0.0018
5-8	0.92288	baseline / auto-like settings	baseline
9	0.92117	svd + cubic	-0.0017
10	0.91522	svd + rank=10	-0.0077
11-13	0.8823	multiquadric kernel	-0.0406
Worst	-0.0066	gaussian kernel	failed

The main observations are:

- Standard `svd` performs better than randomized SVD.
- `rank=20` is the most suitable setting.
- `thin_plate_spline` and auto kernels are the best choices.

- `multiquadric` degrades performance, while `gaussian` fails.

4. cGAN Parameter Search

Nine cGAN settings were first tested for 500 epochs, as listed in Table 3.

Table 3: cGAN search results (500 epochs)

Rank	Avg R^2	Min R^2	Configuration
1	0.9195	0.7525	lr=5e-5, bs=4, adv=0.001
2	0.9165	0.7484	baseline (1e-4, 8, 0.01)
3	0.9165	0.7790	1e-4, 8, 0.001
4	0.9148	0.7409	1e-4, 4, 0.01
5	0.9144	0.7253	5e-4, 8, 0.01
6	0.9133	0.6618	5e-4, 16, 0.001
7	0.9028	0.7628	1e-4, 16, 0.01
8	0.9015	0.7013	1e-4, 8, 0.1
9	0.9012	0.7193	5e-5, 8, 0.01

The trends are clear: smaller learning rate, smaller batch size, and weaker adversarial weight are more suitable in the present small-data regime.

A longer 2000-epoch run further improves cGAN, as shown in Table 4.

Table 4: cGAN full-training result

Stage	Configuration	Avg R^2	Max R^2	Min R^2
Original training	lr=1e-4, bs=8, adv=0.01	0.9209	0.9880	0.7542
Best full training	lr=5e-5, bs=4, adv=0.001	0.9233	0.9884	0.7689

Even after improvement, cGAN still remains below the best POD-RBF configuration.

5. Dataset Size Experiment

The influence of dataset size is summarized in Table 5.

Table 5: Performance under different dataset sizes

Total Samples	Train/Test	POD Avg R^2	cGAN Avg R^2
40	32/8	0.851620	0.795458
60	48/12	0.931414	0.930983
80	64/16	0.943732	0.931468
100	80/20	0.944807	0.892085
120	96/24	0.956387	0.938397
140	112/28	0.943492	0.940818
160	128/32	0.958493	0.951346

Both models improve with more data, but POD-RBF performs better in most settings, especially when the dataset is small. This suggests stronger small-sample efficiency for the reduced-order approach.

6. Conclusion

The current experiments lead to three conclusions:

- **POD-RBF** is the best method in the present setting, with the best configuration being `svd + rank=20 + thin_plate_spline`.
- **cGAN** benefits from tuning and longer training, but still does not surpass POD-RBF.
- As the dataset grows, the gap becomes smaller, but POD-RBF remains the more robust and efficient solution under limited data.